Pulse + Petals

Biomimetic Fashion Technology

Introduction

Wearable technology has traditionally prioritized data display and numerical quantification, using the body's physiological data to display and communicate internal information rather than personal self expression, while fashion couture has relatively stayed separate from technology. This project explores biomimetic kinetic fashion, merging technology, biology, and design into a single expressive system.

Mechanized 3D printed flowers open and close at a speed that is based on the wearer's heart rate. By translating physiological data into visible and blooming expression, the garment creates a beautiful and personal display of internal emotional and physical states.

Purpose

The purpose of this research is to design and create a fully wearable dress that displays the wearer's inner physiological state by visually actuating a gentle, mechanical motion.

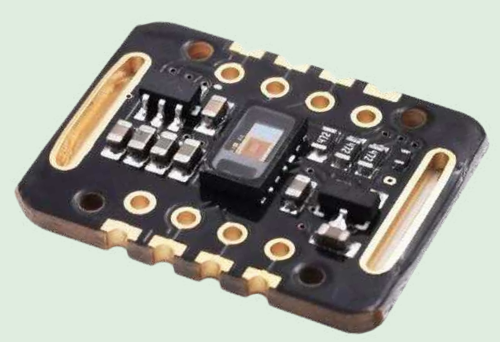
Limitations

- Lack of familiarity with CAD softwares
- MAX30102 sensor accuracy
- Single-User testing
- Mechanism durability
- Availability of 3D printing stations

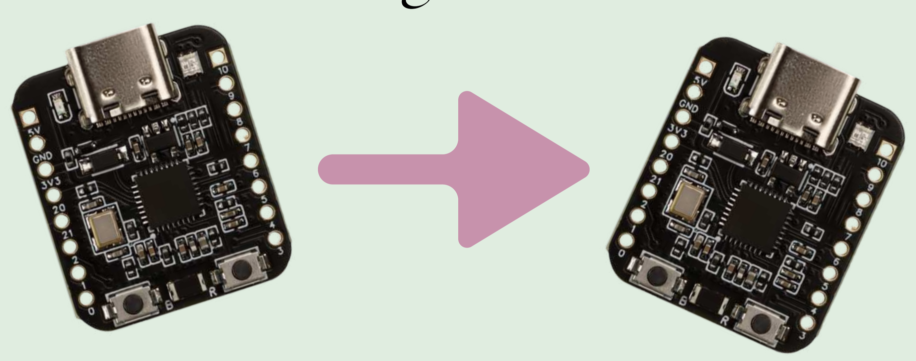
Methods

Mechanical System:
A servo actuates 3D printed petals mimicking flower blooms.

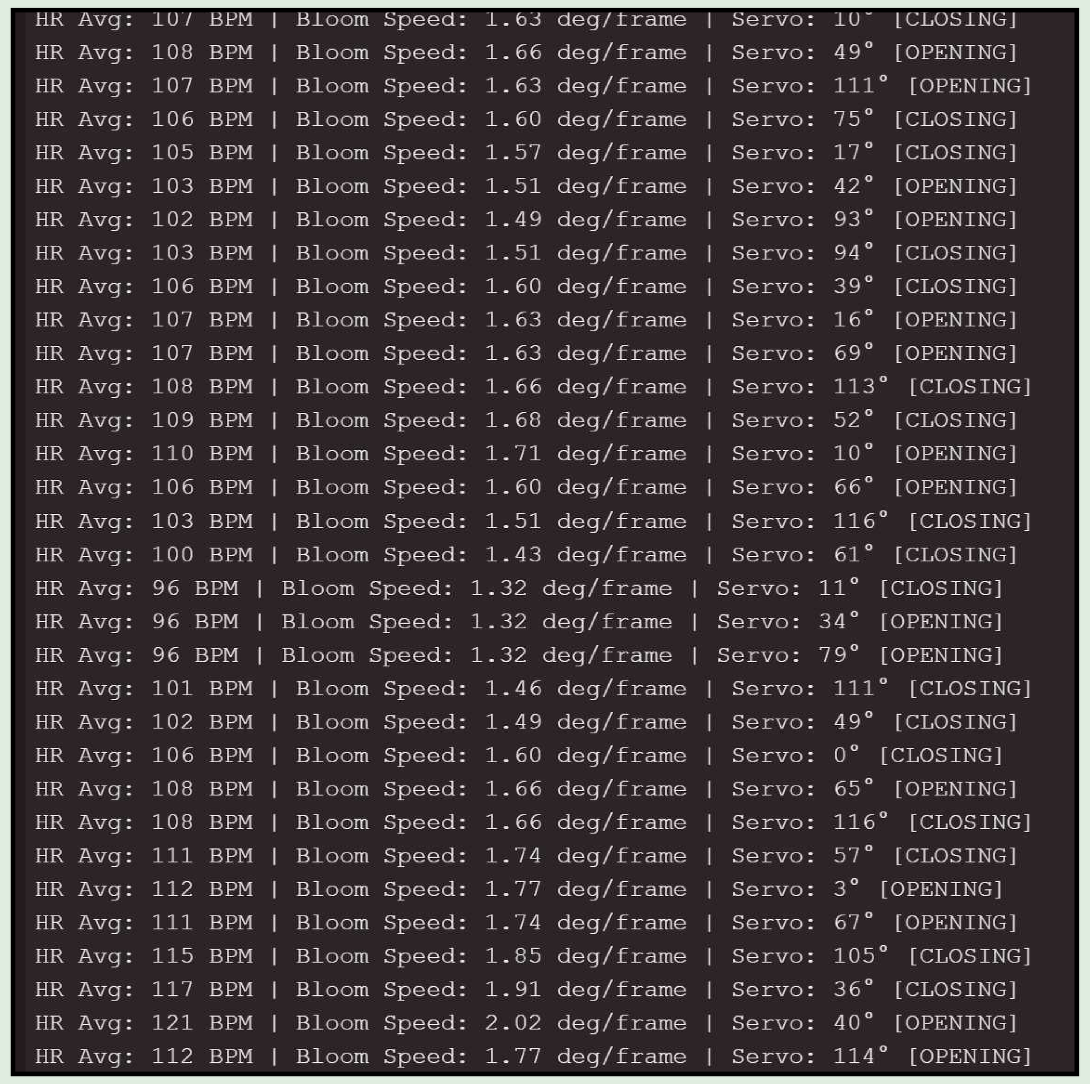
Sensing System:
MAX30102 measuring beats per minute in real-time, taking a scaled average to determine the rate of opening/closing.



Control System:
One sender ESP-32 takes the data from the MAX30102, and transmits it to another ESP-32 via ESPNOW, which processes heart rate data and mapping it to motor control signals.



Results



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HR Avg: 107 BPM | Bloom Speed: 1.63 deg/frame | Servo: 10° [CLOSING]
HR Avg: 108 BPM | Bloom Speed: 1.66 deg/frame | Servo: 49° [OPENING]
HR Avg: 107 BPM | Bloom Speed: 1.63 deg/frame | Servo: 111° [OPENING]
HR Avg: 106 BPM | Bloom Speed: 1.60 deg/frame | Servo: 75° [CLOSING]
HR Avg: 105 BPM | Bloom Speed: 1.57 deg/frame | Servo: 17° [CLOSING]
HR Avg: 103 BPM | Bloom Speed: 1.51 deg/frame | Servo: 42° [OPENING]
HR Avg: 102 BPM | Bloom Speed: 1.49 deg/frame | Servo: 93° [OPENING]
HR Avg: 103 BPM | Bloom Speed: 1.51 deg/frame | Servo: 94° [CLOSING]
HR Avg: 106 BPM | Bloom Speed: 1.60 deg/frame | Servo: 39° [CLOSING]
HR Avg: 107 BPM | Bloom Speed: 1.63 deg/frame | Servo: 16° [OPENING]
HR Avg: 107 BPM | Bloom Speed: 1.63 deg/frame | Servo: 69° [OPENING]
HR Avg: 108 BPM | Bloom Speed: 1.66 deg/frame | Servo: 113° [CLOSING]
HR Avg: 109 BPM | Bloom Speed: 1.68 deg/frame | Servo: 52° [CLOSING]
HR Avg: 110 BPM | Bloom Speed: 1.71 deg/frame | Servo: 10° [OPENING]
HR Avg: 106 BPM | Bloom Speed: 1.60 deg/frame | Servo: 66° [OPENING]
HR Avg: 103 BPM | Bloom Speed: 1.51 deg/frame | Servo: 116° [CLOSING]
HR Avg: 100 BPM | Bloom Speed: 1.43 deg/frame | Servo: 61° [CLOSING]
HR Avg: 96 BPM | Bloom Speed: 1.32 deg/frame | Servo: 11° [CLOSING]
HR Avg: 96 BPM | Bloom Speed: 1.32 deg/frame | Servo: 34° [OPENING]
HR Avg: 96 BPM | Bloom Speed: 1.32 deg/frame | Servo: 79° [OPENING]
HR Avg: 101 BPM | Bloom Speed: 1.46 deg/frame | Servo: 111° [CLOSING]
HR Avg: 102 BPM | Bloom Speed: 1.49 deg/frame | Servo: 49° [CLOSING]
HR Avg: 106 BPM | Bloom Speed: 1.60 deg/frame | Servo: 0° [CLOSING]
HR Avg: 108 BPM | Bloom Speed: 1.66 deg/frame | Servo: 65° [OPENING]
HR Avg: 108 BPM | Bloom Speed: 1.66 deg/frame | Servo: 116° [CLOSING]
HR Avg: 111 BPM | Bloom Speed: 1.74 deg/frame | Servo: 57° [CLOSING]
HR Avg: 112 BPM | Bloom Speed: 1.77 deg/frame | Servo: 3° [OPENING]
HR Avg: 111 BPM | Bloom Speed: 1.74 deg/frame | Servo: 67° [OPENING]
HR Avg: 115 BPM | Bloom Speed: 1.85 deg/frame | Servo: 105° [CLOSING]
HR Avg: 117 BPM | Bloom Speed: 1.91 deg/frame | Servo: 36° [CLOSING]
HR Avg: 121 BPM | Bloom Speed: 2.02 deg/frame | Servo: 40° [OPENING]
HR Avg: 112 BPM | Bloom Speed: 1.77 deg/frame | Servo: 114° [OPENING]
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Figure 1: Serial Output during Heart Rate Testing
Image created by researcher using Arduino, 2026.

Heart rate mapping: Resting (60-80 BPM) = slow blooms, Elevated (100-120 BPM) = rapid movements

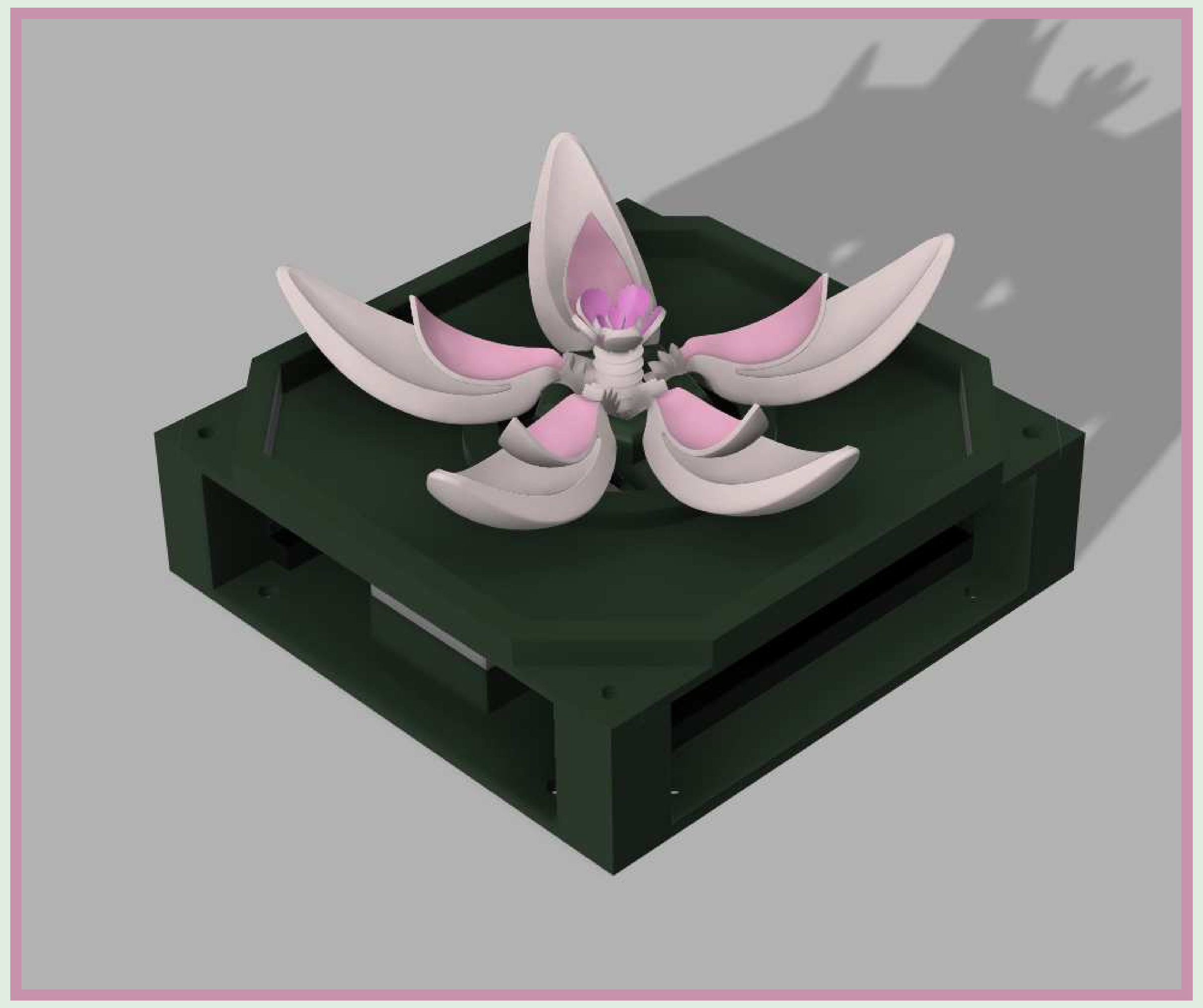

Figure 2: Render of Flower Mechanism
Image created by researcher using Fusion, 2026.

Mechanical Actuator: A rack and pinion system with a custom servo horn and geared stem move the petals.

Conclusion

This prototype demonstrates that physiological data can be effectively translated into kinetic display using biomimetic design. The flower mechanism successfully completed 203 cycles before being limited by servo overload, proving the sensor-to-actuator concept works.

However, limitations in time, printer availability, and software familiarity impacted the project timeline. The bodice structure has been digitally modeled but fabrication remains incomplete. Despite this, the core engineering principles and heart rate integration have been validated. Future work will focus on completing the bodice construction and refining integration.

Future Research

Power optimization: Explore battery options for running without a computer for a wearable amount of time.

Design refinement: Develop and engineer a wearable bodice primarily for comfort and aesthetics.

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